

RUSTON, LA, USA

WMO: 722251

Lat: 32.514N

Lon: 92.588W

Elev: 311

StdP: 14.53

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 53958

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.6	27.7	10.5	9.5	35.2	16.4	12.7	39.3	18.6	61.5	17.0	57.8	2.4	20	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	98.9	75.0	96.2	75.1	93.4	75.2	80.0	88.7	79.1	87.9	78.3	86.9	5.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.1	147.8	82.5	77.2	143.0	82.1	76.3	138.8	81.7	43.7	89.1	42.7	87.8	41.9	87.2	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.8	14.3	12.2	DB	18.2	100.7	4.7	2.9	14.8	102.8	12.0	104.6	9.4	106.2	6.0	108.4
			WB	17.6	81.9	4.5	1.8	14.3	83.2	11.7	84.3	9.2	85.3	6.0	86.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	46.5	50.6	57.6	63.9	72.2	79.8	81.9	81.9	76.9	65.7	54.8	49.2
	DBStd	15.02	10.45	10.87	9.69	7.39	6.32	4.23	4.39	4.46	5.59	8.46	9.52	9.99
	HDD50	559	196	116	40	5	0	0	0	0	0	3	58	141
	HDD65	2316	580	418	264	106	20	0	0	0	1	98	328	501
	CDD50	6094	86	133	277	422	689	893	990	988	808	491	201	116
	CDD65	2378	6	15	36	74	244	443	525	523	359	121	21	11
	CDH74	23559	15	69	257	633	2091	4577	5725	5581	3340	1108	123	40
	CDH80	10715	0	7	35	101	732	2159	2889	2863	1540	371	16	2
(14) Wind	WSAvg	4.3	5.0	5.5	5.5	5.3	4.5	3.9	3.4	3.1	2.9	3.7	4.3	4.9
(15) Precipitation	PrecAvg	56.5	5.7	4.9	5.1	5.2	4.8	4.4	3.9	3.4	3.9	4.4	4.6	5.3
	PrecMax	74.1	14.9	8.3	9.4	18.2	10.3	12.0	9.4	7.7	9.7	21.7	12.0	9.8
	PrecMin	37.4	0.6	0.7	1.1	2.3	0.9	1.0	1.1	0.9	0.5	0.4	1.2	0.5
	PrecStd	10.0	3.3	2.5	2.2	3.5	2.6	2.6	2.0	1.8	2.3	4.7	2.6	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.3	81.0	84.4	85.9	90.9	99.6	100.3	103.9	97.7	91.3	82.5	78.7
		MCWB	63.5	65.1	67.0	68.6	72.5	73.0	75.8	77.4	73.6	70.4	69.7	70.2
	2%	DB	72.1	75.4	80.2	82.3	88.8	96.9	98.7	99.0	94.7	87.9	77.4	73.9
		MCWB	63.6	64.9	64.9	66.6	72.1	74.0	75.5	76.0	73.6	70.5	66.2	67.4
	5%	DB	68.1	71.9	77.0	81.0	87.1	92.8	95.7	95.1	91.5	84.4	73.3	70.3
		MCWB	61.2	62.7	63.3	65.8	71.5	75.5	76.2	75.3	74.5	69.9	65.3	65.4
	10%	DB	63.8	68.1	73.0	78.2	84.5	90.4	92.9	92.4	89.5	80.8	70.5	67.1
		MCWB	57.9	61.4	62.3	64.7	70.6	74.9	76.0	75.4	74.2	67.9	64.0	63.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.6	69.3	70.9	73.5	77.2	80.4	80.9	81.8	79.8	76.7	72.5	72.0
		MCDB	72.2	75.7	76.5	79.9	84.8	88.0	90.2	89.5	87.7	82.4	77.9	76.8
	2%	WB	65.9	67.5	68.4	71.4	75.4	79.0	79.8	80.2	78.3	75.1	70.5	69.2
		MCDB	70.4	72.2	75.4	77.4	83.4	87.1	89.3	88.5	86.2	82.2	75.0	72.6
	5%	WB	63.4	65.4	66.6	69.8	74.2	78.0	79.0	79.0	77.1	73.4	68.3	67.1
		MCDB	65.5	70.4	73.2	75.4	82.1	85.8	88.4	87.7	85.0	79.9	72.1	69.7
	10%	WB	59.4	63.0	65.1	67.9	73.0	77.1	78.2	78.0	75.8	71.7	65.5	63.7
		MCDB	62.6	67.4	71.0	74.3	80.6	84.8	87.4	86.9	84.1	76.6	69.7	65.9
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	20.4	22.3	22.6	20.7	19.9	19.8	20.3	21.5	23.8	22.9	19.6
		MCDBR	23.4	23.1	24.2	24.4	22.1	23.2	23.0	23.6	24.0	25.5	23.8	22.0
		MCWBR	15.5	13.6	11.8	11.0	7.9	6.6	5.5	6.3	7.6	10.8	14.0	15.3
	5% WB	MCDBR	19.7	18.5	19.3	19.1	18.7	18.5	19.4	19.6	19.6	20.2	18.4	19.3
		MCWBR	15.7	13.1	11.7	11.8	8.1	7.0	6.3	7.1	7.6	10.6	13.7	15.7
(40) Clear-Sky Solar Irradiance	taub		0.325	0.340	0.364	0.402	0.430	0.472	0.487	0.484	0.428	0.364	0.346	0.329
	taud		2.525	2.481	2.419	2.307	2.285	2.212	2.196	2.202	2.333	2.481	2.485	2.516
	Ebn at Noon		285	291	290	282	273	261	256	255	266	278	274	276
	Edh at Noon		27	31	36	41	43	46	46	45	38	30	28	26
(44) All-Sky Solar Radiation	RadAvg		848	965	1362	1701	1886	2001	1972	1846	1589	1297	947	739
	RadStd		82	164	112	121	126	141	131	155	148	182	102	94

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page